

# “Comandante”: Braving the Waves With Near Real-Time Virtual Production Workflows

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## ABSTRACT

The naval feature film “Comandante” is to be shot on a water stage within a waterproof but very low resolution LED volume. We developed our Near Real-Time (NRT) workflow to immediately improve the quality of a shot, seamlessly bridging In-Camera VFX (ICVFX) and post-production, allowing VFX to start on-set as soon as the director says ‘cut’. The NRT workflow brings together key advancements in lens calibration, machine learning and real-time rendering to deliver higher quality composites of what was just shot to the filmmakers, in a matter of minutes.

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## 1 INTRODUCTION

“Comandante” tells the story of Salvatore Todaro, an Italian naval submarine officer who in 1940 saved the lives of the enemy sailors whose steamboats he was tasked with sinking. To bring this unique humanitarian tale of war-time compassion to life, the filmmakers opted to shoot on a water tank within an LED volume, using rich dynamic lighting to realize both the grand scale of the open seas, as well as the action and emotion heavy moments between sailors on the submarine’s conning tower.

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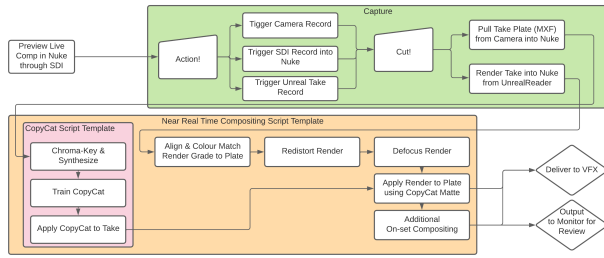
**Figure 1: Example shot from our recent test at 80six using ROE Carbon CB3 LED panels, a severe example of moiré illustrating the need for the NRT screen replacement.**

High resolution LED panels such as ROE Ruby or Black Pearl are not designed to get wet, so lower resolution, water resistant outdoor LED panels must be used instead. These panels are unsuitable for use in In-Camera VFX (ICVFX) due to the severe moiré patterns they create. This created the challenge and opportunity:

- (1) Can we keep the rich, dynamic light cast on the talent, water and physical set, and replace the section of LED seen through the camera with a higher quality version?
- (2) And can we do this quickly enough on-set, so that the director and DP can be confident in what they shot and can move on to the next set-up? i.e. can it happen in *Near Real-Time*?

## 2 DESIGN & IMPLEMENTATION

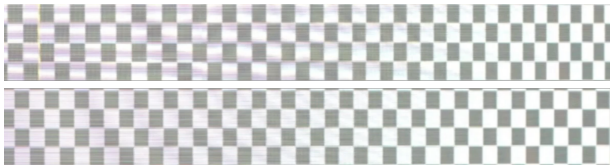
The goal is to augment what was shot with a higher quality render, and to do this as quickly as possible. The key components to this process are illustrated in Fig. 2, with core contributions outlined below.



**Figure 2: Near Real-Time workflow diagram, showing the steps from capture to composite, and then to review.**

*Rendering & Solving Latency.* Epic Games’ “Unreal” is used to deliver perspective-correct real-time renders of our nautical scene to the LED wall sections in real-time. For each take, the camera tracking position and lens metadata is captured into a Sequence in Unreal. Foundry’s “UnrealReader” tool is then used to render the images offline from the virtual camera, ready to be composited.

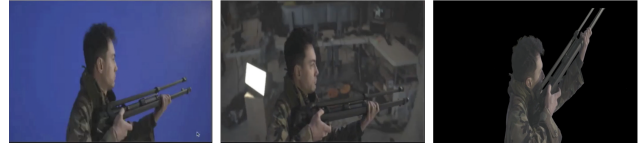
A benefit of this is the mitigation of latency or visual lag caused by the delay between tracking, rendering and delivery to the LED wall, which can be up to many frames. By re-rendering using the true recorded values of the tracked camera’s position, any lag between tracking and real-time render is removed, allowing filmmakers complete freedom over their shot movements.



**Figure 3: Comparing lens undistortion quality. Top, a close-up of a grid undistorted using Foundry’s “LensDistortion” tool. Bottom, undistorting by applying Cooke’s model “out of the box”. This result is astonishing: the aspect ratios of the checkers are perfectly consistent across the entire grid.**

*Lens Calibration & Metadata.* Central to marrying the real and virtual is modelling the physical lens, allowing us to distort the Unreal render (or undistort the live action plate.) Instead of shooting lens grids, we worked with Cooke Optics to use their extremely accurate factory lens mapping, which embeds distortion and shading parameters into each lens itself [Cooke Optics 2022]. Once extracted we were able to get highly accurate distortion maps, even as the lens “breathes”. See Fig. 3 for a comparison in mapping quality.

*Machine Learning for Virtual Production.* Before compositing, the last step is to isolate our LED background to be replaced. We developed a simple and quick method of creating a per-show and per-shot dataset that is used to train an ML model to matte our foreground talent from the background, see Fig. 4. The dataset preparation can happen in advance during pre-light, test or rehearsal stages, with training taking from tens of minutes to a couple of hours.



**Figure 4: ML for Postviz. Left, the actor rehearses in front of the LED wall set to blue. Centre, the chroma-keyed footage is composited onto synthetic backgrounds to create a data set. Right, after training Foundry’s “CopyCat” ML tool, the LED wall can be removed in the real take, ready to be replaced.**



**Figure 5: Examples of NRT compositing. Left, a high-quality composite of foreground image onto the replaced rendered background. Right, the same system replacing the floor as well as the wall, allowing complete freedom of shooting.**

It is now the compositor’s job to get the filmmakers what they need to see to feel confident that they shot what they wanted. Example composites from our tests can be seen in Fig. 5.

### 3 DISCUSSION

The NRT workflow allows the use of lower res LED panels without the moiré artefacts usually associated with them. This offers Productions the option to use more cost effective panels, or higher contrast outdoor panels, and keep the richness of the LED light.

The NRT workflow helps bridge the gap between on-set and post. Using the recorded camera tracks and advanced lens metadata with the UnrealReader, both on-set or offline VFX becomes easier. Also, by re-rendering the LED wall with true, timecode-accurate camera motion, latency in the VP system is mitigated.

Finally, the NRT workflow shows how ML can be used to solve complicated problems on-set, such as matting. The same process could be used to perform other tasks such as rig-removal, “face swap”-like performance transfer, digital beauty work, and more.

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Cooke Optics. 2022. LMF Metadata Specification. [https://bitbucket.org/cookeoptics/cookelensmetadata/src/master/Cooke\\_Metadata\\_File\\_Specification.md](https://bitbucket.org/cookeoptics/cookelensmetadata/src/master/Cooke_Metadata_File_Specification.md).